```python
import os
import h5py
import time
import matplotlib
import numpy as np
import pandas as pd
import seaborn as sns
from pandas import DataFrame
import matplotlib.pyplot as plt
from matplotlib import gridspec
import os
import h5py
import time
import matplotlib
import matplotlib.ticker as ticker
import numpy as np
import pandas as pd
import seaborn as sns
from pandas import DataFrame
import matplotlib.pyplot as plt
from matplotlib import gridspec
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score, mean_squared_error
from sklearn.metrics import classification_report
from sklearn.pipeline import Pipeline
%matplotlib inline
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
from sklearn.preprocessing import StandardScaler
from sklearn.preprocessing import MinMaxScaler
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score, mean_squared_error
from sklearn.metrics import classification_report
from sklearn.pipeline import Pipeline
import keras
from keras.models import Sequential
from keras.layers import Dense
from keras.layers import Dropout
from deap import algorithms, base, creator, tools
import tensorflow as tf
import tensorflow as tf
from tensorflow.keras import Sequential
from tensorflow.keras.layers import LSTM, Dense, Dropout, Masking,
TimeDistributed
from keras.models import Sequential
from keras.layers import Dense, Conv1D
```

```python
from keras.layers import Dropout, Input
from keras import initializers
from keras.layers import MaxPooling1D
from keras.models import Model
from keras.layers import TimeDistributed, Flatten
from keras.layers import Dense, Activation, Dropout
from keras.callbacks import EarlyStopping
from keras.callbacks import ModelCheckpoint
%matplotlib inline

filename = 'N-CMAPSS_DS05.h5'

# Time tracking, Operation time (min):  0.003
t = time.process_time()

# Load data
with h5py.File(filename, 'r') as hdf:
        # Development set
        W_dev = np.array(hdf.get('W_dev'))             # W
        X_s_dev = np.array(hdf.get('X_s_dev'))         # X_s
        X_v_dev = np.array(hdf.get('X_v_dev'))         # X_v
        T_dev = np.array(hdf.get('T_dev'))             # T
        Y_dev = np.array(hdf.get('Y_dev'))             # RUL
        A_dev = np.array(hdf.get('A_dev'))             # Auxiliary

        # Test set
        W_test = np.array(hdf.get('W_test'))           # W
        X_s_test = np.array(hdf.get('X_s_test'))       # X_s
        X_v_test = np.array(hdf.get('X_v_test'))       # X_v
        T_test = np.array(hdf.get('T_test'))           # T
        Y_test = np.array(hdf.get('Y_test'))           # RUL
        A_test = np.array(hdf.get('A_test'))           # Auxiliary

        # Varnams
        W_var = np.array(hdf.get('W_var'))
        X_s_var = np.array(hdf.get('X_s_var'))
        X_v_var = np.array(hdf.get('X_v_var'))
        T_var = np.array(hdf.get('T_var'))
        A_var = np.array(hdf.get('A_var'))

        # from np.array to list dtype U4/U5
        W_var = list(np.array(W_var, dtype='U20'))
        X_s_var = list(np.array(X_s_var, dtype='U20'))
        X_v_var = list(np.array(X_v_var, dtype='U20'))
        T_var = list(np.array(T_var, dtype='U20'))
        A_var = list(np.array(A_var, dtype='U20'))

W = np.concatenate((W_dev, W_test), axis=0)
X_s = np.concatenate((X_s_dev, X_s_test), axis=0)
X_v = np.concatenate((X_v_dev, X_v_test), axis=0)
```

```
T = np.concatenate((T_dev, T_test), axis=0)
Y = np.concatenate((Y_dev, Y_test), axis=0)
A = np.concatenate((A_dev, A_test), axis=0)

print('')
print("Operation time (min): " , (time.process_time()-t)/60)
print('')
print ("W shape: " + str(W.shape))
print ("X_s shape: " + str(X_s.shape))
print ("X_v shape: " + str(X_v.shape))
print ("Y shape: " + str(Y.shape))
print ("T shape: " + str(T.shape))
print ("A shape: " + str(A.shape))
```

```
Operation time (min):  0.04661458333333334

W shape: (6912652, 4)
X_s shape: (6912652, 14)
X_v shape: (6912652, 14)
Y shape: (6912652, 1)
T shape: (6912652, 10)
A shape: (6912652, 4)
```

```
W = np.concatenate((W_dev, W_test), axis=0)
X_s = np.concatenate((X_s_dev, X_s_test), axis=0)
Y = np.concatenate((Y_dev, Y_test), axis=0)
A = np.concatenate((A_dev, A_test), axis=0)

X = np.concatenate((W, X_s, A, Y), axis=1)
Y=  np.concatenate((A, Y), axis=1)
Y = DataFrame(data=Y, columns=A_var+['Y'])
X = DataFrame(data=X, columns=W_var+X_s_var+A_var+['Y'])

Y.describe()
```

```
                 unit         cycle            Fc            hs
Y
count  6.912652e+06  6.912652e+06  6.912652e+06  6.912652e+06
6.912652e+06
mean   5.239374e+00  4.002810e+01  2.226115e+00  2.958030e-01
3.900481e+01
std    2.904136e+00  2.368351e+01  8.078304e-01  4.564029e-01
2.376696e+01
min    1.000000e+00  1.000000e+00  1.000000e+00  0.000000e+00
0.000000e+00
25%    2.000000e+00  2.000000e+01  2.000000e+00  0.000000e+00
1.900000e+01
50%    6.000000e+00  3.900000e+01  2.000000e+00  0.000000e+00
3.800000e+01
```

```
75%    8.000000e+00  5.900000e+01  3.000000e+00  1.000000e+00
5.800000e+01
max    1.000000e+01  1.000000e+02  3.000000e+00  1.000000e+00
9.900000e+01
```

```
Y.drop("cycle",axis=1,inplace=True)
Y.drop("Fc",axis=1,inplace=True)
Y.drop("hs",axis=1,inplace=True)
Y.describe()
```

```
               unit             Y
count  6.912652e+06  6.912652e+06
mean   5.239374e+00  3.900481e+01
std    2.904136e+00  2.376696e+01
min    1.000000e+00  0.000000e+00
25%    2.000000e+00  1.900000e+01
50%    6.000000e+00  3.800000e+01
75%    8.000000e+00  5.800000e+01
max    1.000000e+01  9.900000e+01
```

```
X.drop("cycle",axis=1,inplace=True)
X.drop("Fc",axis=1,inplace=True)
X.drop("hs",axis=1,inplace=True)
X.drop("alt",axis=1,inplace=True)
X.describe()
```

```
               Mach           TRA            T2           T24
T30  \
count  6.912652e+06  6.912652e+06  6.912652e+06  6.912652e+06
6.912652e+06
mean   5.319760e-01  5.996256e+01  4.909356e+02  5.699286e+02
1.334474e+03
std    1.208898e-01  1.834395e+01  1.985279e+01  2.112542e+01
6.869315e+01
min    3.150000e-04  2.355452e+01  4.369255e+02  4.998511e+02
1.091806e+03
25%    4.393620e-01  4.631803e+01  4.741667e+02  5.546204e+02
1.287931e+03
50%    5.371380e-01  6.257768e+01  4.960515e+02  5.678197e+02
1.329839e+03
75%    6.330240e-01  7.664008e+01  5.072438e+02  5.838195e+02
1.373982e+03
max    7.492590e-01  8.876890e+01  5.343834e+02  6.341967e+02
1.540888e+03

                T48           T50           P15            P2
P21  \
count  6.912652e+06  6.912652e+06  6.912652e+06  6.912652e+06
6.912652e+06
mean   1.638378e+03  1.129818e+03  1.304474e+01  1.019378e+01
```

```
1.324339e+01
std    1.239105e+02  6.251462e+01  2.883393e+00  2.421477e+00
2.927303e+00
min    1.095354e+03  7.923581e+02  6.700688e+00  5.285918e+00
6.802729e+00
25%    1.552724e+03  1.085702e+03  1.049740e+01  7.972020e+00
1.065726e+01
50%    1.647608e+03  1.120127e+03  1.338949e+01  1.060965e+01
1.359339e+01
75%    1.711727e+03  1.165307e+03  1.528072e+01  1.209400e+01
1.551342e+01
max    1.993206e+03  1.356282e+03  2.044899e+01  1.568410e+01
2.076040e+01

                P24          Ps30           P40           P50
Nf  \
count  6.912652e+06  6.912652e+06  6.912652e+06  6.912652e+06
6.912652e+06
mean   1.606218e+01  2.379398e+02  2.422144e+02  1.019697e+01
1.953364e+03
std    3.450926e+00  5.919650e+01  6.003612e+01  2.758136e+00
1.873247e+02
min    8.186232e+00  9.248497e+01  9.466065e+01  4.635623e+00
1.471000e+03
25%    1.319101e+01  1.929589e+02  1.965787e+02  7.665725e+00
1.835894e+03
50%    1.617596e+01  2.256779e+02  2.301263e+02  1.051082e+01
1.997333e+03
75%    1.847904e+01  2.724938e+02  2.773639e+02  1.232305e+01
2.112084e+03
max    2.643985e+01  4.485121e+02  4.553945e+02  1.672457e+01
2.285550e+03

                 Nc            Wf          unit             Y
count  6.912652e+06  6.912652e+06  6.912652e+06  6.912652e+06
mean   8.248141e+03  2.548930e+00  5.239374e+00  3.900481e+01
std    2.277543e+02  7.877672e-01  2.904136e+00  2.376696e+01
min    7.429764e+03  5.963687e-01  1.000000e+00  0.000000e+00
25%    8.094517e+03  1.980253e+00  2.000000e+00  1.900000e+01
50%    8.237001e+03  2.340848e+00  6.000000e+00  3.800000e+01
75%    8.383625e+03  2.948869e+00  8.000000e+00  5.800000e+01
max    8.921408e+03  5.607345e+00  1.000000e+01  9.900000e+01

df_A = DataFrame(data=A, columns=A_var)
df_A.describe()

               unit         cycle            Fc            hs
count  6.912652e+06  6.912652e+06  6.912652e+06  6.912652e+06
mean   5.239374e+00  4.002810e+01  2.226115e+00  2.958030e-01
std    2.904136e+00  2.368351e+01  8.078304e-01  4.564029e-01
```

```
min     1.000000e+00  1.000000e+00  1.000000e+00  0.000000e+00
25%     2.000000e+00  2.000000e+01  2.000000e+00  0.000000e+00
50%     6.000000e+00  3.900000e+01  2.000000e+00  0.000000e+00
75%     8.000000e+00  5.900000e+01  3.000000e+00  1.000000e+00
max     1.000000e+01  1.000000e+02  3.000000e+00  1.000000e+00
```

```
print('Engine units in df: ', np.unique(X['unit']))
print('Engine units in df: ', np.unique(Y['unit']))
```

```
Engine units in df:  [ 1.  2.  3.  4.  5.  6.  7.  8.  9. 10.]
Engine units in df:  [ 1.  2.  3.  4.  5.  6.  7.  8.  9. 10.]
```

```
X_train= X.loc[(X.unit == 1) | (X.unit == 3) | (X.unit == 5) | (X.unit
== 6) | (X.unit == 7) | (X.unit == 9) | (X.unit == 10) ]
Y_train=Y.loc[(Y.unit == 1) | (Y.unit == 3) | (Y.unit == 5) |  (Y.unit
== 6) | (Y.unit == 7) | (Y.unit == 9) | (Y.unit == 10) ]
X_test=X.loc[(X.unit == 2) | (X.unit == 4) | (X.unit == 8) ]
Y_test=Y.loc[(Y.unit == 2) | (Y.unit == 4) | (Y.unit == 8) ]
```

```
print('Engine units in df: ', np.unique(X_train['unit']))
print('Engine units in df: ', np.unique(Y_train['unit']))
print('Engine units in df: ', np.unique(X_test['unit']))
print('Engine units in df: ', np.unique(Y_test['unit']))
print("X_train size is ",X_train.shape)
print("X_test size is ",X_test.shape)
print("y_train size is ",Y_train.shape)
print("y_test size is ",Y_test.shape)
```

```
Engine units in df:  [ 1.  3.  5.  6.  7.  9. 10.]
Engine units in df:  [ 1.  3.  5.  6.  7.  9. 10.]
Engine units in df:  [2. 4. 8.]
Engine units in df:  [2. 4. 8.]
X_train size is  (4900033, 19)
X_test size is  (2012619, 19)
y_train size is  (4900033, 2)
y_test size is  (2012619, 2)
```

```
X_train.drop("unit",axis=1,inplace=True)
X_test.drop("unit",axis=1,inplace=True)
X_train.drop("Y",axis=1,inplace=True)
X_test.drop("Y",axis=1,inplace=True)
Y_train.drop("unit",axis=1,inplace=True)
Y_test.drop("unit",axis=1,inplace=True)
print("X_train size is ",X_train.shape)
print("X_test size is ",X_test.shape)
print("y_train size is ",Y_train.shape)
print("y_test size is ",Y_test.shape)
```

```
X_train size is  (4900033, 17)
X_test size is  (2012619, 17)
```

```
y_train size is  (4900033, 1)
y_test size is  (2012619, 1)
```

```
C:\Users\Kamal\AppData\Local\Temp\ipykernel_36196\816249269.py:1:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation:
https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#
returning-a-view-versus-a-copy
  X_train.drop("unit",axis=1,inplace=True)
C:\Users\Kamal\AppData\Local\Temp\ipykernel_36196\816249269.py:2:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation:
https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#
returning-a-view-versus-a-copy
  X_test.drop("unit",axis=1,inplace=True)
C:\Users\Kamal\AppData\Local\Temp\ipykernel_36196\816249269.py:3:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation:
https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#
returning-a-view-versus-a-copy
  X_train.drop("Y",axis=1,inplace=True)
C:\Users\Kamal\AppData\Local\Temp\ipykernel_36196\816249269.py:4:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation:
https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#
returning-a-view-versus-a-copy
  X_test.drop("Y",axis=1,inplace=True)
C:\Users\Kamal\AppData\Local\Temp\ipykernel_36196\816249269.py:5:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation:
https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#
returning-a-view-versus-a-copy
  Y_train.drop("unit",axis=1,inplace=True)
C:\Users\Kamal\AppData\Local\Temp\ipykernel_36196\816249269.py:6:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation:
https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#
```

```
returning-a-view-versus-a-copy
  Y_test.drop("unit",axis=1,inplace=True)
```

```
def reshape_time_series(data, sequence_length):
    """
    Reshapes time series data for CNN input.

    Parameters:
    data (np.array): The time series dataset, expected to be a 2D
array.
    sequence_length (int): The length of the input sequences for the
CNN.

    Returns:
    np.array: Reshaped dataset suitable for CNN input.
    """
    X = []
    for i in range(len(data) - sequence_length):
        X.append(data[i:i + sequence_length])

    return np.array(X)

X_train = reshape_time_series(X_train,20)
X_test = reshape_time_series(X_test,20)
Y_train = reshape_time_series(Y_train,20)
Y_test = reshape_time_series(Y_test,20)

from keras.models import Sequential
from keras.layers import Conv1D, MaxPooling1D
from keras.layers import LSTM
from keras.layers import Dense, Dropout, Flatten

print("X_train size is ",X_train.shape)
print("X_test size is ",X_test.shape)
print("y_train size is ",Y_train.shape)
print("y_test size is ",Y_test.shape)
```

```
X_train size is  (4900013, 20, 17)
X_test size is  (2012599, 20, 17)
y_train size is  (4900013, 20, 1)
y_test size is  (2012599, 20, 1)
```

```
from sklearn.model_selection import train_test_split

def split_dataset(X, y, train_size=0.8):
    """
    Splits the dataset into training and validation sets.

    Parameters:
    X (np.array): Features of the dataset.
    y (np.array): Labels or targets of the dataset.
```

```
    train_size (float): Proportion of the dataset to include in the
training set.

    Returns:
    X_train, X_val, y_train, y_val: Split dataset into training and
validation sets.
    """
    X_train, X_val, y_train, y_val = train_test_split(X, y,
train_size=train_size, random_state=42)
    return X_train, X_val, y_train, y_val

X_train, X_val, Y_train, Y_val = split_dataset(X_train, Y_train,
train_size=0.8)

print("X_train shape:", X_train.shape)
print("y_train shape:", Y_train.shape)
print("X_val shape:", X_val.shape)
print("y_val shape:", Y_val.shape)
```

```
X_train shape: (3920010, 20, 17)
y_train shape: (3920010, 20, 1)
X_val shape: (980003, 20, 17)
y_val shape: (980003, 20, 1)
```

```
def reshape_labels_for_regression(y_data):
    """
    Reshapes the labels for a regression task.

    Parameters:
    y_data (np.array): Original labels with shape (number_of_samples,
20, 10).

    Returns:
    np.array: Reshaped labels with shape (number_of_samples, 1).
    """
    # Assuming we want to predict the first value at the last time
step of each sequence
    # Adjust the index '0' if a different value is desired
    y_reshaped = y_data[:, -1, 0]  # Select the last time step, first
value
    return y_reshaped.reshape(-1, 1)

import kerastuner as kt
from tensorflow.keras.models import Model
from tensorflow.keras.layers import Input, Dense, LSTM, Conv1D,
MaxPooling1D, Flatten, Dropout, Concatenate, Masking
from tensorflow.keras.optimizers import Adam
from tensorflow.keras import regularizers
from tensorflow.keras.callbacks import EarlyStopping, ModelCheckpoint,
ReduceLROnPlateau
from sklearn.model_selection import GroupShuffleSplit
```

```
import copy
import os
from tensorflow.keras.models import load_model
```

```
C:\Users\Kamal\AppData\Local\Temp\ipykernel_36196\1420853871.py:1:
DeprecationWarning: `import kerastuner` is deprecated, please use
`import keras_tuner`.
  import kerastuner as kt
```

```
sequence_length = 20
num_features =  17
class GATuner(kt.tuners.RandomSearch):
    def run_trial(self, trial, *args, **kwargs):
        hp = trial.hyperparameters
        model = self.hypermodel.build(hp)

        # Extract the validation data
        fit_args = copy.deepcopy(args)
        fit_kwargs = copy.deepcopy(kwargs)
        val_data = fit_kwargs.pop('validation_data', None)
        epochs = fit_kwargs.pop('epochs', None)

        # Fit the model
        if val_data:
            history = model.fit(*fit_args, validation_data=val_data,
epochs=epochs, **fit_kwargs)
        else:
            history = model.fit(*fit_args, epochs=epochs,
**fit_kwargs)

        # Evaluate the model
        val_loss = history.history['val_loss'][-1]
        self.oracle.update_trial(trial.trial_id, {'val_loss':
val_loss})
        self.save_model(trial.trial_id, model)
    def save_model(self, trial_id, model, step=0):
        """Saves the model for a given trial."""
        fname = os.path.join(self.get_trial_dir(trial_id), 'model.h5')
        try:
            model.save(fname)
            print(f"Model saved to {fname}")  # Adding logging
        except Exception as e:
            print(f"Error saving model: {e}")  # Error handling

# Define the hypermodel
def build_model(hp):
    sequence_length = 20
    num_features = 17

    input_cnn = Input(shape=(sequence_length, num_features))
```

```
    input_lstm = Input(shape=(sequence_length, num_features))

    # CNN branch with tunable parameters
    cnn_branch = Masking(mask_value=-99.)(input_cnn)
    for i in range(hp.Int('cnn_layers', 1, 3)):
        cnn_branch = Conv1D(filters=hp.Int(f'cnn_filters_{i}', 32,
128, step=32),

kernel_size=hp.Choice(f'cnn_kernel_size_{i}', [3, 5]),
                            padding='same',

activation=hp.Choice(f'cnn_activation_{i}', ['relu', 'tanh']))
(cnn_branch)
        if i < hp.Int('cnn_layers', 1, 3) - 1:
            cnn_branch = MaxPooling1D(pool_size=2, padding='same')
(cnn_branch)
    cnn_branch = Flatten()(cnn_branch)
    cnn_branch = Dense(hp.Int('cnn_dense_units', 64, 256, step=64),
                       activation='relu')(cnn_branch)
    cnn_branch = Dropout(hp.Float('cnn_dropout', 0.2, 0.5))
(cnn_branch)

    # LSTM branch with tunable parameters
    lstm_branch = Masking(mask_value=-99.)(input_lstm)
    for j in range(hp.Int('lstm_layers', 1, 3)):
        lstm_branch = LSTM(units=hp.Int(f'lstm_units_{j}', 32, 128,
step=32),
                           return_sequences=j < hp.Int('lstm_layers',
1, 3) - 1,
                           activation='tanh')(lstm_branch)
    lstm_branch = Flatten()(lstm_branch)
    lstm_branch = Dense(hp.Int('lstm_dense_units', 64, 256, step=64),
                        activation='relu')(lstm_branch)
    lstm_branch = Dropout(hp.Float('lstm_dropout', 0.2, 0.5))
(lstm_branch)

    # Concatenating the outputs of the two branches
    merged = Concatenate()([cnn_branch, lstm_branch])
    merged = Dense(hp.Int('merged_dense_units', 64, 256, step=64),
activation='relu')(merged)
    merged = Dropout(hp.Float('merged_dropout', 0.2, 0.5))(merged)

    # Output layer
    output = Dense(1, activation='linear')(merged)

    # Create and compile the model
    model = Model(inputs=[input_cnn, input_lstm], outputs=output)
    model.compile(optimizer=Adam(hp.Float('learning_rate', 1e-4, 1e-2,
sampling='log')),
                  loss='mean_squared_error')
```

```
    return model

# Instantiate the GA tuner
ga_tuner = GATuner(
    hypermodel=build_model,
    objective='val_loss',
    max_trials=5,
    executions_per_trial=1,
    directory='ga_tuning_NewData',
    project_name='ga_tuning_project1'
)

# Set up callbacks
es = EarlyStopping(monitor='val_loss', mode='min', verbose=1,
patience=15)
mc = ModelCheckpoint('best_deeper_model.h5', monitor='val_loss',
mode='min', verbose=1, save_best_only=True)
reduce_lr = ReduceLROnPlateau(monitor='val_loss', factor=0.5,
patience=5, min_lr=0.00001, verbose=1)

ga_tuner.search([X_train, X_train], Y_train,
                epochs=30,
                validation_data=([X_val, X_val], Y_val),
                batch_size=32,
                callbacks=[es, mc, reduce_lr])

best_model = ga_tuner.get_best_models(num_models=1)[0]
```

```
Reloading Tuner from ga_tuning_NewData\ga_tuning_project1\tuner0.json

Search: Running Trial #3

Value             |Best Value So Far |Hyperparameter
2                 |2                 |cnn_layers
64                |128               |cnn_filters_0
5                 |3                 |cnn_kernel_size_0
tanh              |relu              |cnn_activation_0
256               |64                |cnn_dense_units
0.31229           |0.49881           |cnn_dropout
1                 |2                 |lstm_layers
96                |64                |lstm_units_0
192               |128               |lstm_dense_units
0.46974           |0.29373           |lstm_dropout
64                |256               |merged_dense_units
0.27698           |0.41036           |merged_dropout
0.0016771         |0.00020286        |learning_rate
96                |32                |cnn_filters_1
5                 |3                 |cnn_kernel_size_1
tanh              |relu              |cnn_activation_1
```

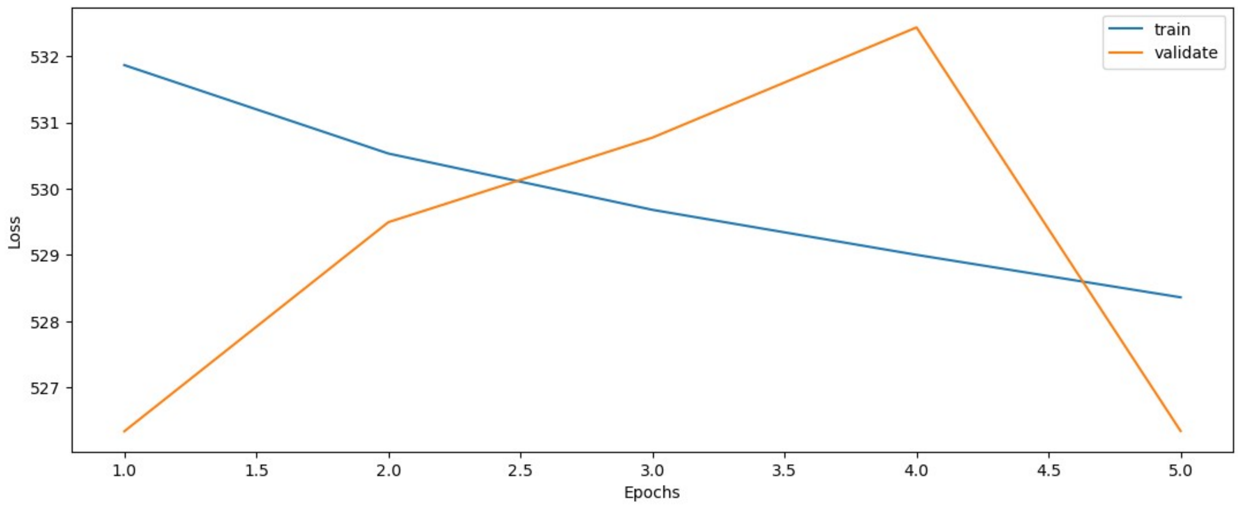
```
96                |32               |lstm_units_1


Traceback (most recent call last):
  File "c:\Users\Kamal\anaconda3\envs\myenv\lib\site-packages\
keras_tuner\src\engine\base_tuner.py", line 273, in
_try_run_and_update_trial
    self._run_and_update_trial(trial, *fit_args, **fit_kwargs)
  File "c:\Users\Kamal\anaconda3\envs\myenv\lib\site-packages\
keras_tuner\src\engine\base_tuner.py", line 238, in
_run_and_update_trial
    results = self.run_trial(trial, *fit_args, **fit_kwargs)
  File "C:\Users\Kamal\AppData\Local\Temp\
ipykernel_36196\1818877440.py", line 16, in run_trial
    history = model.fit(*fit_args, validation_data=val_data,
epochs=epochs, **fit_kwargs)
  File "c:\Users\Kamal\anaconda3\envs\myenv\lib\site-packages\keras\
src\utils\traceback_utils.py", line 70, in error_handler
    raise e.with_traceback(filtered_tb) from None
  File "c:\Users\Kamal\anaconda3\envs\myenv\lib\site-packages\
tensorflow\python\framework\constant_op.py", line 98, in
convert_to_eager_tensor
    return ops.EagerTensor(value, ctx.device_name, dtype)
numpy.core._exceptions._ArrayMemoryError: Unable to allocate 4.97 GiB
for an array with shape (3920010, 20, 17) and data type float32

----------------------------------------------------------------------
-----
RuntimeError                              Traceback (most recent call
last)
c:\Users\Kamal\Documents\AUC\thesiswork\N-Cmapps\Trial5 - Copy.ipynb
Cell 22 line 9
     <a
href='vscode-notebook-cell:/c%3A/Users/Kamal/Documents/AUC/thesiswork/
N-Cmapps/Trial5%20-%20Copy.ipynb#X26sZmlsZQ%3D%3D?line=92'>93</a> mc =
ModelCheckpoint('best_deeper_model.h5', monitor='val_loss',
mode='min', verbose=1, save_best_only=True)
     <a
href='vscode-notebook-cell:/c%3A/Users/Kamal/Documents/AUC/thesiswork/
N-Cmapps/Trial5%20-%20Copy.ipynb#X26sZmlsZQ%3D%3D?line=93'>94</a>
reduce_lr = ReduceLROnPlateau(monitor='val_loss', factor=0.5,
patience=5, min_lr=0.00001, verbose=1)
---> <a
href='vscode-notebook-cell:/c%3A/Users/Kamal/Documents/AUC/thesiswork/
N-Cmapps/Trial5%20-%20Copy.ipynb#X26sZmlsZQ%3D%3D?line=95'>96</a>
ga_tuner.search([X_train, X_train], Y_train,
     <a
href='vscode-notebook-cell:/c%3A/Users/Kamal/Documents/AUC/thesiswork/
N-Cmapps/Trial5%20-%20Copy.ipynb#X26sZmlsZQ%3D%3D?line=96'>97</a>
epochs=30,
```

```
     <a
href='vscode-notebook-cell:/c%3A/Users/Kamal/Documents/AUC/thesiswork/
N-Cmapps/Trial5%20-%20Copy.ipynb#X26sZmlsZQ%3D%3D?line=97'>98</a>
validation_data=([X_val, X_val], Y_val),
     <a
href='vscode-notebook-cell:/c%3A/Users/Kamal/Documents/AUC/thesiswork/
N-Cmapps/Trial5%20-%20Copy.ipynb#X26sZmlsZQ%3D%3D?line=98'>99</a>
batch_size=32,
    <a
href='vscode-notebook-cell:/c%3A/Users/Kamal/Documents/AUC/thesiswork/
N-Cmapps/Trial5%20-%20Copy.ipynb#X26sZmlsZQ%3D%3D?line=99'>100</a>
callbacks=[es, mc, reduce_lr])
    <a
href='vscode-notebook-cell:/c%3A/Users/Kamal/Documents/AUC/thesiswork/
N-Cmapps/Trial5%20-%20Copy.ipynb#X26sZmlsZQ%3D%3D?line=101'>102</a>
best_model = ga_tuner.get_best_models(num_models=1)[0]

File c:\Users\Kamal\anaconda3\envs\myenv\lib\site-packages\
keras_tuner\src\engine\base_tuner.py:234, in BaseTuner.search(self,
*fit_args, **fit_kwargs)
    232     self.on_trial_begin(trial)
    233     self._try_run_and_update_trial(trial, *fit_args,
**fit_kwargs)
--> 234     self.on_trial_end(trial)
    235 self.on_search_end()

File c:\Users\Kamal\anaconda3\envs\myenv\lib\site-packages\
keras_tuner\src\engine\base_tuner.py:338, in
BaseTuner.on_trial_end(self, trial)
    332 def on_trial_end(self, trial):
    333     """Called at the end of a trial.
    334 
    335     Args:
    336         trial: A `Trial` instance.
    337     """
--> 338     self.oracle.end_trial(trial)
    339     self.save()

File c:\Users\Kamal\anaconda3\envs\myenv\lib\site-packages\
keras_tuner\src\engine\oracle.py:108, in
synchronized.<locals>.wrapped_func(*args, **kwargs)
    106     LOCKS[oracle].acquire()
    107     THREADS[oracle] = thread_name
--> 108 ret_val = func(*args, **kwargs)
    109 if need_acquire:
    110     THREADS[oracle] = None

File c:\Users\Kamal\anaconda3\envs\myenv\lib\site-packages\
keras_tuner\src\engine\oracle.py:586, in Oracle.end_trial(self, trial)
    584 if not self._retry(trial):
```

```
    585     self.end_order.append(trial.trial_id)
--> 586     self._check_consecutive_failures()
    588 self._save_trial(trial)
    589 self.save()

File c:\Users\Kamal\anaconda3\envs\myenv\lib\site-packages\
keras_tuner\src\engine\oracle.py:543, in
Oracle._check_consecutive_failures(self)
    541     consecutive_failures = 0
    542 if consecutive_failures == self.max_consecutive_failed_trials:
--> 543     raise RuntimeError(
    544         "Number of consecutive failures exceeded the limit "
    545         f"of {self.max_consecutive_failed_trials}.\n"
    546         + (trial.message or "")
    547     )

RuntimeError: Number of consecutive failures exceeded the limit of 3.
Traceback (most recent call last):
  File "c:\Users\Kamal\anaconda3\envs\myenv\lib\site-packages\
keras_tuner\src\engine\base_tuner.py", line 273, in
_try_run_and_update_trial
    self._run_and_update_trial(trial, *fit_args, **fit_kwargs)
  File "c:\Users\Kamal\anaconda3\envs\myenv\lib\site-packages\
keras_tuner\src\engine\base_tuner.py", line 238, in
_run_and_update_trial
    results = self.run_trial(trial, *fit_args, **fit_kwargs)
  File "C:\Users\Kamal\AppData\Local\Temp\
ipykernel_36196\1818877440.py", line 16, in run_trial
    history = model.fit(*fit_args, validation_data=val_data,
epochs=epochs, **fit_kwargs)
  File "c:\Users\Kamal\anaconda3\envs\myenv\lib\site-packages\keras\
src\utils\traceback_utils.py", line 70, in error_handler
    raise e.with_traceback(filtered_tb) from None
  File "c:\Users\Kamal\anaconda3\envs\myenv\lib\site-packages\
tensorflow\python\framework\constant_op.py", line 98, in
convert_to_eager_tensor
    return ops.EagerTensor(value, ctx.device_name, dtype)
numpy.core._exceptions._ArrayMemoryError: Unable to allocate 4.97 GiB
for an array with shape (3920010, 20, 17) and data type float32
```

```python
def plot_loss(fit_history):
    plt.figure(figsize=(13,5))
    plt.plot(range(1, len(fit_history.history['loss'])+1),
fit_history.history['loss'], label='train')
    plt.plot(range(1, len(fit_history.history['val_loss'])+1),
fit_history.history['val_loss'], label='validate')
    plt.xlabel('Epochs')
    plt.ylabel('Loss')
    plt.legend()
```

```
    plt.show()

plot_loss(history)
```




```
def evaluate(y_true, y_hat, label='test'):
    mse = mean_squared_error(y_true, y_hat)
    rmse = np.sqrt(mse)
    variance = r2_score(y_true, y_hat)
    print('{} set RMSE:{}, R2:{}'.format(label, rmse, variance))


y_hat_test = model.predict(X_test)
print (y_hat_test)
```

```
62894/62894 [==============================] - 287s 5ms/step
[[36.85976]
 [36.85976]
 [36.85976]
 ...
 [36.85976]
 [36.85976]
 [36.85976]]
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call
last)
c:\Users\Kamal\Documents\AUC\thesiswork\N-Cmapps\Trial5.ipynb Cell 24
line 3
      <a
href='vscode-notebook-cell:/c%3A/Users/Kamal/Documents/AUC/thesiswork/
N-Cmapps/Trial5.ipynb#X34sZmlsZQ%3D%3D?line=0'>1</a> y_hat_test =
model.predict(X_test)
      <a
```

```
href='vscode-notebook-cell:/c%3A/Users/Kamal/Documents/AUC/thesiswork/
N-Cmapps/Trial5.ipynb#X34sZmlsZQ%3D%3D?line=1'>2</a> print
(y_hat_test)
----> <a
href='vscode-notebook-cell:/c%3A/Users/Kamal/Documents/AUC/thesiswork/
N-Cmapps/Trial5.ipynb#X34sZmlsZQ%3D%3D?line=2'>3</a> evaluate(Y_test,
y_hat_test)

c:\Users\Kamal\Documents\AUC\thesiswork\N-Cmapps\Trial5.ipynb Cell 24
line 2
      <a
href='vscode-notebook-cell:/c%3A/Users/Kamal/Documents/AUC/thesiswork/
N-Cmapps/Trial5.ipynb#X34sZmlsZQ%3D%3D?line=0'>1</a> def
evaluate(y_true, y_hat, label='test'):
----> <a
href='vscode-notebook-cell:/c%3A/Users/Kamal/Documents/AUC/thesiswork/
N-Cmapps/Trial5.ipynb#X34sZmlsZQ%3D%3D?line=1'>2</a>     mse =
mean_squared_error(y_true, y_hat)
      <a
href='vscode-notebook-cell:/c%3A/Users/Kamal/Documents/AUC/thesiswork/
N-Cmapps/Trial5.ipynb#X34sZmlsZQ%3D%3D?line=2'>3</a>     rmse =
np.sqrt(mse)
      <a
href='vscode-notebook-cell:/c%3A/Users/Kamal/Documents/AUC/thesiswork/
N-Cmapps/Trial5.ipynb#X34sZmlsZQ%3D%3D?line=3'>4</a>     variance =
r2_score(y_true, y_hat)

File c:\Users\Kamal\anaconda3\envs\myenv\lib\site-packages\sklearn\
metrics\_regression.py:442, in mean_squared_error(y_true, y_pred,
sample_weight, multioutput, squared)
    382 def mean_squared_error(
    383     y_true, y_pred, *, sample_weight=None,
multioutput="uniform_average", squared=True
    384 ):
    385     """Mean squared error regression loss.
    386
    387     Read more in the :ref:`User Guide <mean_squared_error>`.
   (...)
    440     0.825...
    441     """
--> 442     y_type, y_true, y_pred, multioutput = _check_reg_targets(
    443         y_true, y_pred, multioutput
    444     )
    445     check_consistent_length(y_true, y_pred, sample_weight)
    446     output_errors = np.average((y_true - y_pred) ** 2, axis=0,
weights=sample_weight)

File c:\Users\Kamal\anaconda3\envs\myenv\lib\site-packages\sklearn\
metrics\_regression.py:101, in _check_reg_targets(y_true, y_pred,
multioutput, dtype)
```

```
     67 """Check that y_true and y_pred belong to the same regression
task.
     68
     69 Parameters
   (...)
     98     correct keyword.
     99 """
    100 check_consistent_length(y_true, y_pred)
--> 101 y_true = check_array(y_true, ensure_2d=False, dtype=dtype)
    102 y_pred = check_array(y_pred, ensure_2d=False, dtype=dtype)
    104 if y_true.ndim == 1:

File c:\Users\Kamal\anaconda3\envs\myenv\lib\site-packages\sklearn\
utils\validation.py:915, in check_array(array, accept_sparse,
accept_large_sparse, dtype, order, copy, force_all_finite, ensure_2d,
allow_nd, ensure_min_samples, ensure_min_features, estimator,
input_name)
    910     raise ValueError(
    911         "dtype='numeric' is not compatible with arrays of
bytes/strings."
    912         "Convert your data to numeric values explicitly
instead."
    913     )
    914 if not allow_nd and array.ndim >= 3:
--> 915     raise ValueError(
    916         "Found array with dim %d. %s expected <= 2."
    917         % (array.ndim, estimator_name)
    918     )
    920 if force_all_finite:
    921     _assert_all_finite(
    922         array,
    923         input_name=input_name,
    924         estimator_name=estimator_name,
    925         allow_nan=force_all_finite == "allow-nan",
    926     )

ValueError: Found array with dim 3. None expected <= 2.
```

```python
print (y_hat_test.shape)
print (Y_test.shape)
reshaped_data = Y_test[:, 0, 0]
reshaped_data = reshaped_data[:, np.newaxis]
print(reshaped_data.shape)
evaluate(reshaped_data, y_hat_test)
```

```
(2012599, 1)
(2012599, 20, 1)
(2012599, 1)
test set RMSE:26.011789063409275, R2:-0.05579286571155184
```

```
df = pd.DataFrame(reshaped_data)
df.to_csv('output.csv', index=False)
```